

RECIPE SHARING PLATFORM

Machine Evaluation Submission

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Technology Stack: MongoDB, Express.js, React.js, Node.js

1. INTRODUCTION

The Recipe Sharing Platform is a full-stack MERN application that allows users to register, log in securely, create recipes, browse recipes, and update their own recipes.

The application demonstrates:

- JWT Authentication
- Password Hashing using bcrypt
- MongoDB Database Management
- REST API Development
- React Frontend Development
- External API Integration

2. PROJECT OBJECTIVES

- Build a secure authentication system.
- Create REST APIs using Express.js.
- Manage recipe data using MongoDB.
- Develop a responsive React frontend.
- Integrate external image APIs.
- Demonstrate complete MERN stack proficiency.

3. TECHNOLOGY STACK

Frontend

- React.js
- Axios
- Bootstrap
- React Router

Backend

- Node.js
- Express.js

Database

- MongoDB

Authentication

- JWT
- Bcryptjs(password hashing)

External API

- TheMealDB API

4. SYSTEM ARCHITECTURE

User



React Frontend



REST API (Express.js)



MongoDB Database

Workflow:

1. User sends request.
2. React calls API.
3. Express processes request.
4. MongoDB stores/retrieves data.
5. Response returned to frontend.

5. DATABASE DESIGN

User Schema

Fields:

- Name
- Email
- Password (Hashed)

Recipe Schema

Fields:

- Title
- Ingredients
- Instructions
- User Reference

Relationship:

One User → Many Recipes

6. JWT AUTHENTICATION FLOW

Registration

↓

Password Hashing

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Database Storage

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Login

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JWT Token Generation

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Token Stored in Browser

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Protected Route Access

Benefits:

- Secure Authentication
- Stateless Sessions
- Route Protection

7. API ENDPOINTS

Authentication APIs

POST /api/auth/register

POST /api/auth/login

Recipe APIs

POST /api/recipes

GET /api/recipes

GET /api/recipes/:id

PUT /api/recipes/:id

8. FRONTEND IMPLEMENTATION

Login Page

Features:

- Email Login
- Password Login
- JWT Storage

Registration Page

Features:

- User Registration
- Validation

Dashboard

Features:

- Recipe Listing
- Responsive Design

Create Recipe

Features:

- Add Title
- Add Ingredients
- Add Instructions

Recipe Details

Features:

- View Recipe
- External Image Display

9. EXTERNAL API INTEGRATION

API Used:

TheMealDB

Purpose:

Display recipe images based on recipe title.

Example:

<https://www.themealdb.com/api/json/v1/1/search.php?s=Chicken>

Error Handling:

- Default image shown when API fails.
- User-friendly error messages.

10. SECURITY FEATURES

Password Hashing

bcryptjs used for secure password storage.

JWT Authentication

Protects private routes.

Authorization

Only recipe owner can update recipes.

Environment Variables

Sensitive data stored in .env.

Validation

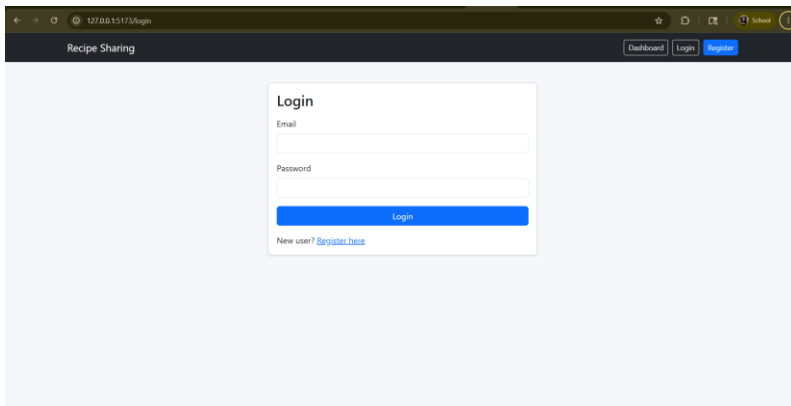
Every fields are validated

11. TESTING

- Completed Testing:
- User Registration
- User Login
- JWT Verification
- Recipe Creation
- Recipe Listing
- Recipe Update
- API Integration
- Error Handling

12. APPLICATION SCREENSHOTS

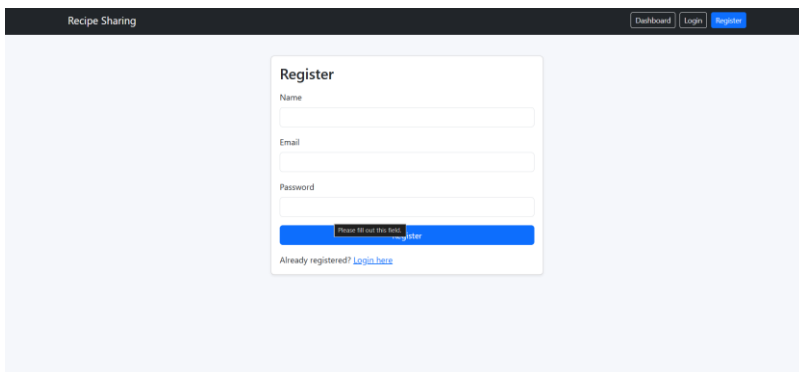
Screenshot 1 – Login Page



The screenshot shows a web browser window with the URL '127.0.0.1:5173/login'. The page title is 'Recipe Sharing'. In the top right corner, there are links for 'Dashboard', 'Login', and 'Register'. The main content area features a 'Login' form with two input fields for 'Email' and 'Password', a blue 'Login' button, and a link for 'New user? Register here'.

The login page allows registered users to authenticate using email and password.

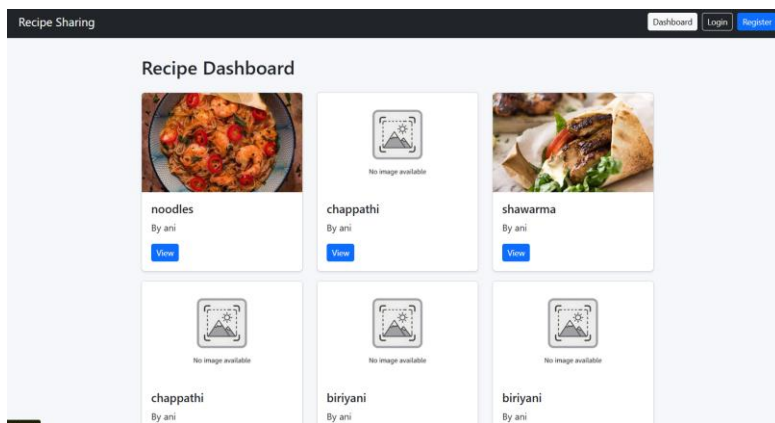
Screenshot 2 – Registration Page



The screenshot shows a web browser window with the URL '127.0.0.1:5173/register'. The page title is 'Recipe Sharing'. In the top right corner, there are links for 'Dashboard', 'Login', and 'Register'. The main content area features a 'Register' form with three input fields for 'Name', 'Email', and 'Password', a blue 'Register' button, and a link for 'Already registered? Login here'.

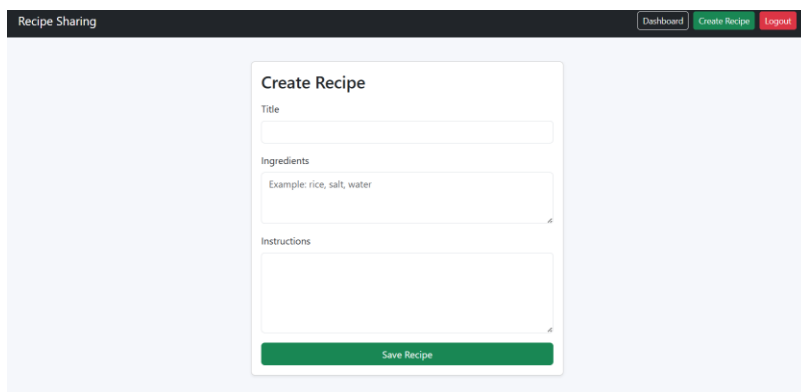
Users can create a new account using the registration form.

Screenshot 3 – Dashboard Page



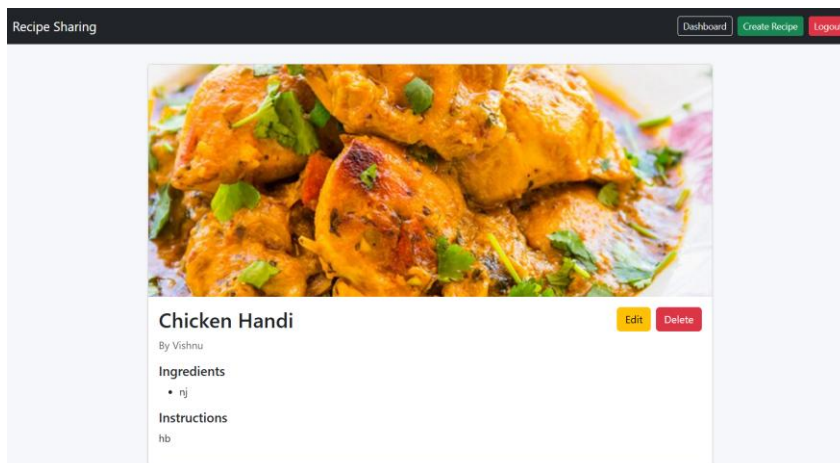
Displays all available recipes fetched from the backend API.

Screenshot 4 – Create Recipe Page



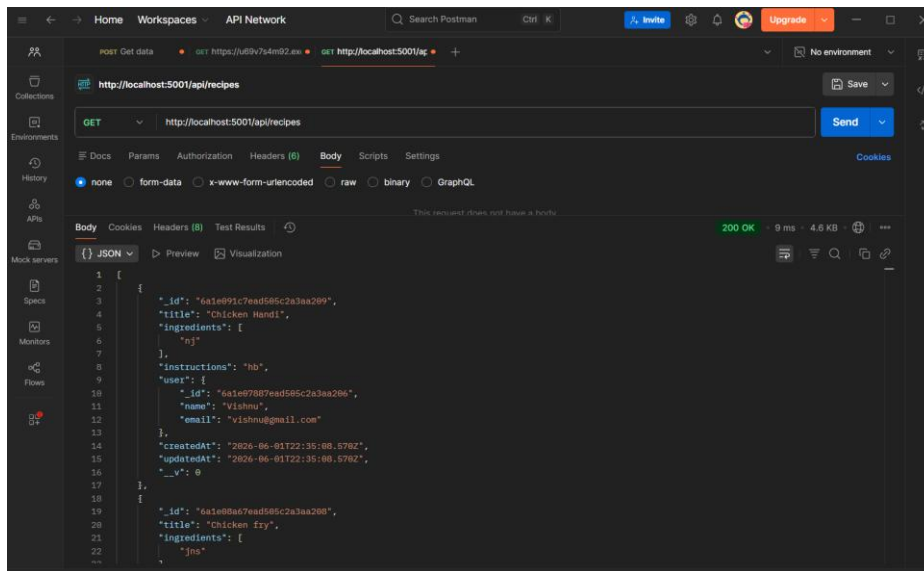
Allows authenticated users to create and submit recipes.

Screenshot 5 – Recipe Details Page



Displays recipe information and image fetched from external API.

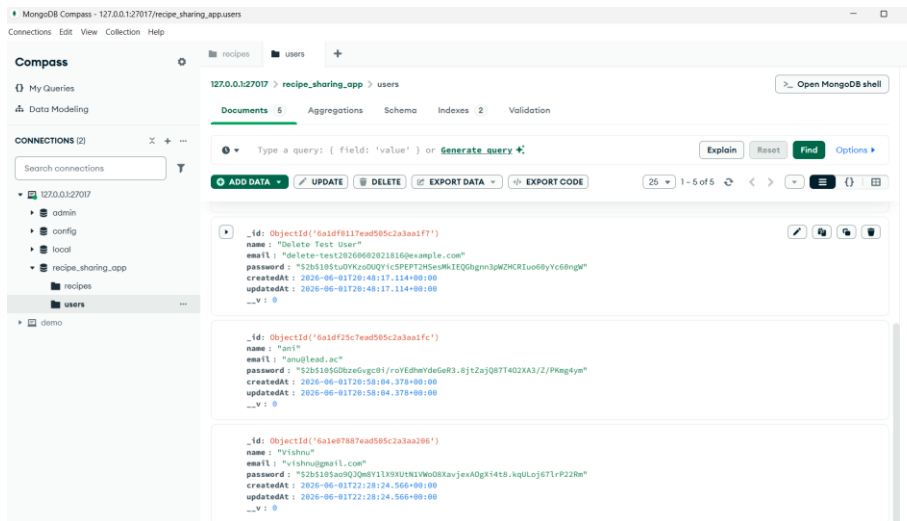
Screenshot 8 – API Response Testing



Shows successful API responses from backend

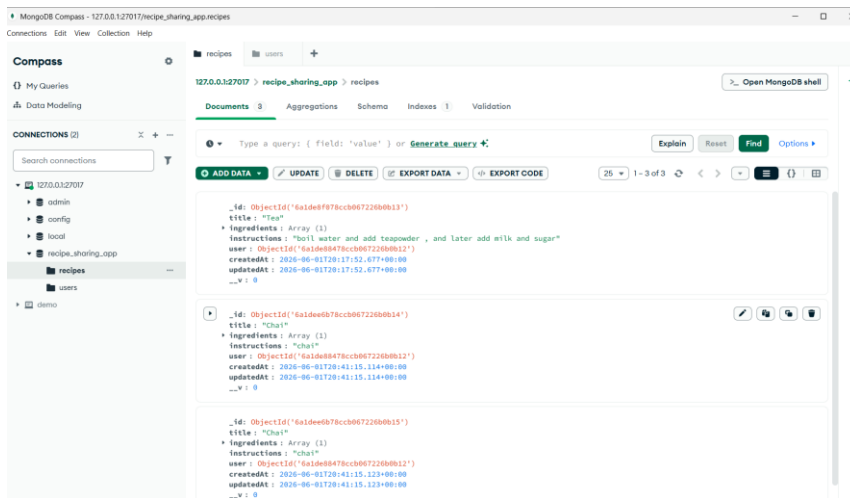
13. MONGODB DATABASE SCREENSHOTS

Users Collection



Stores user account information.

Recipes Collection



Stores recipe information and references to users.

MongoDB Compass Database View

[INSERT DATABASE OVERVIEW SCREENSHOT HERE]

Description:

Displays collections and stored documents.

14. GITHUB REPOSITORY

GitHub Repository Link:

<https://github.com/vishnucax/recipe-sharing-app>

Repository Contents:

- Frontend Source Code
- Backend Source Code
- Documentation
- README

15. HOW TO RUN THE PROJECT

Prerequisites

- Node.js
- MongoDB
- MongoDB Compass
- Git

Clone Repository

git clone <https://github.com/vishnucax/recipe-sharing-app.git>

Backend Setup

cd backend

npm install

npm run dev

Frontend Setup

cd frontend

npm install

npm start

Environment Variables

Create .env file:

MONGO_URI=your_mongodb_connection_string

JWT_SECRET=your_secret_key

PORT=5000

17. FUTURE ENHANCEMENTS

- Recipe Categories
- User Profiles
- Comments System
- Recipe Likes
- Image Upload Feature
- Admin Dashboard

18. CONCLUSION

The Recipe Sharing Platform successfully demonstrates full-stack MERN development by implementing authentication, database management, REST APIs, frontend development, and third-party API integration. The project showcases practical software engineering skills and industry-standard development practices.